Year 10 Set 4 · Foundation · Lesson 1

Straight-line graphs: finding the gradient

w/c 19 October 2026 · OnMaths · Created by Mr Ali

Today I will learn to...

Goals

1. Find the gradient of a line from a graph

2. Find the gradient from two coordinates

Do now: retrieval practice

4 mins

1. Work out 12 ÷ 4

   3

2. Work out 7 − (−2)

   9

3. Work out −6 ÷ 2

   −3

4. Write the point 3 right and 2 up from the origin

   (3, 2)

Objective 1

Find the gradient of a line from a graph

Key idea

Learn

!

The gradient tells you how steep a line is. Pick two points on the line. Gradient = up ÷ across. If the line goes DOWN from left to right, the gradient is negative.

Gradient =

$$\frac{\text{up}}{\text{across}}$$

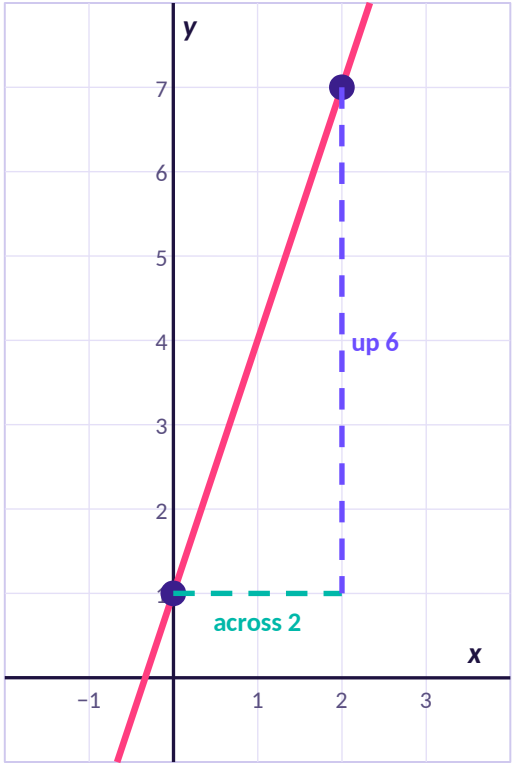
Objective 1: Find the gradient of a line from a graph

I do: example 1

I do

Find the gradient of the line.

1. Pick two points on the line: (0, 1) and (2, 7)
2. Up: 7 − 1 = 6
3. Across: 2 − 0 = 2
4. Gradient = 6 ÷ 2 = 3

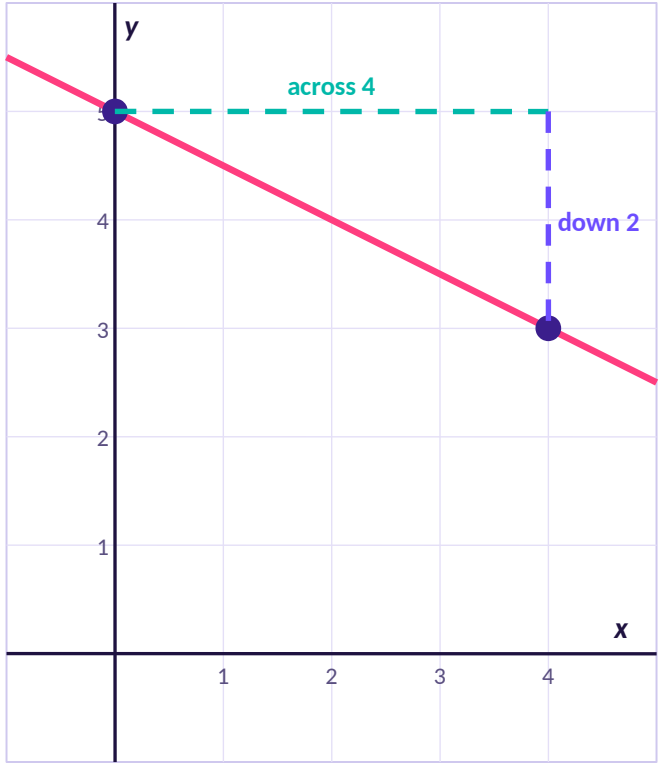


I do: example 2

I do

Find the gradient of the line.

1. Pick two points on the line: (0, 5) and (4, 3)
2. The line goes down, so the gradient is negative
3. Down 2, across 4
4. Gradient = −2 ÷ 4 = −½

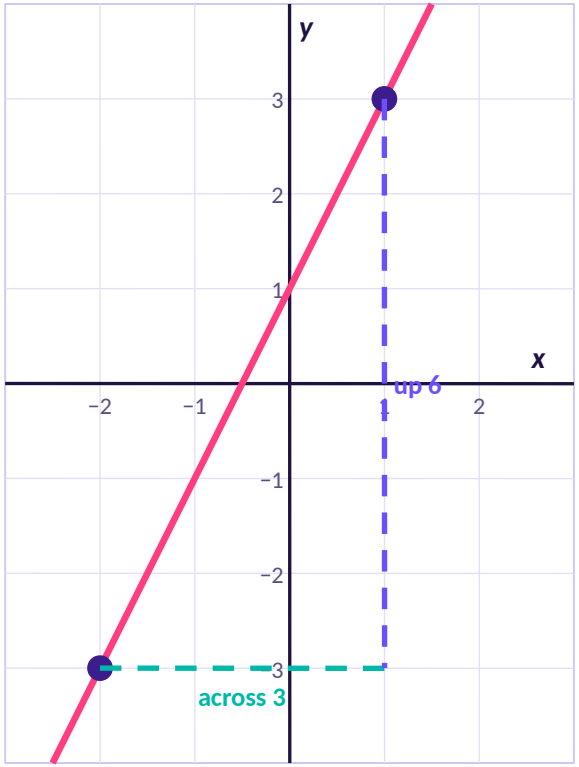


I do: example 3

I do

Find the gradient of the line.

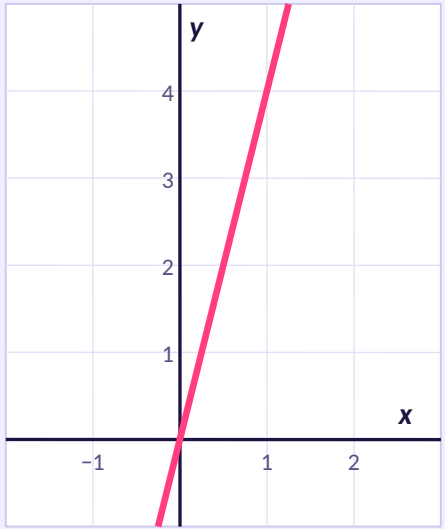
1. Pick two points on the line: (−2, −3) and (1, 3)
2. Up: 3 − (−3) = 6
3. Across: 1 − (−2) = 3
4. Gradient = 6 ÷ 3 = 2



Mini whiteboards

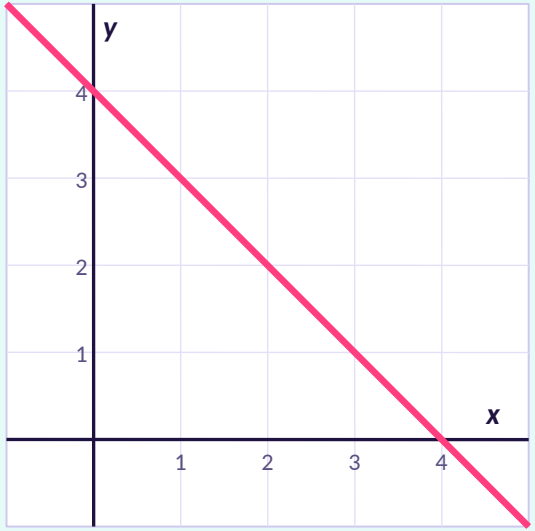
We do

1. Find the gradient of this line.



Answer: 4

2. Find the gradient of this line.

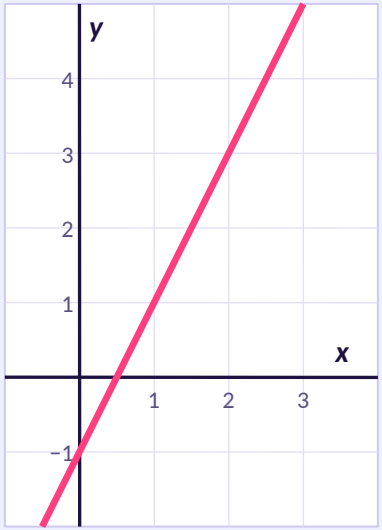


Answer: −1

Mini whiteboards (continued)

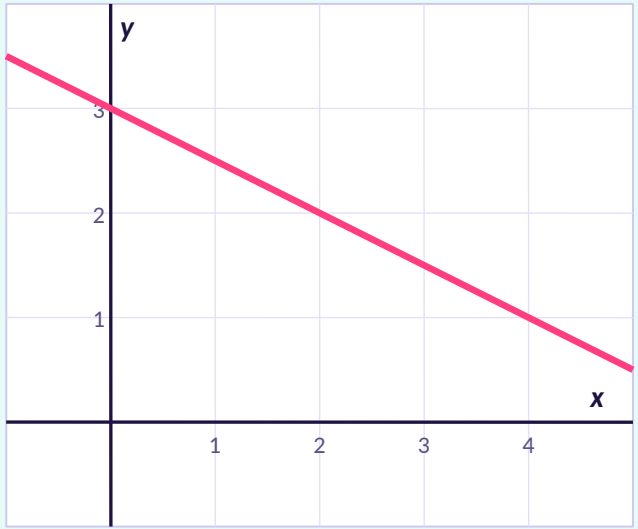
We do

3. Find the gradient of this line.



Answer: 2

4. Find the gradient of this line.



Answer: −½

You do: choose your challenge

You do

Bronze

A line goes up 6 squares for every 2 squares across. Find the gradient.

Silver

A line goes down 4 squares for every 2 squares across. Find the gradient.

Gold

A line goes up 3 squares for every 6 squares across. Give the gradient as a fraction.

Platinum

Two lines start at (0, 0). Line A reaches (2, 6) and line B reaches (3, 6). Which is steeper, and by how much?

You do: answers

Check

Bronze

$6 \div 2 = 3$

Silver

$-4 \div 2 = -2$

Gold

$3 \div 6 = \frac{1}{2}$

Platinum

A: $6 \div 2 = 3$. B: $6 \div 3 = 2$. Line A is steeper by 1

If you finish early

Extra

Bronze

A line goes up 8 squares for every 4 squares across. Find the gradient.

2

Silver

A line goes down 9 squares for every 3 squares across. Find the gradient.

−3

Gold

A line goes up 2 squares for every 8 squares across. Give the gradient as a fraction.

¼

Platinum

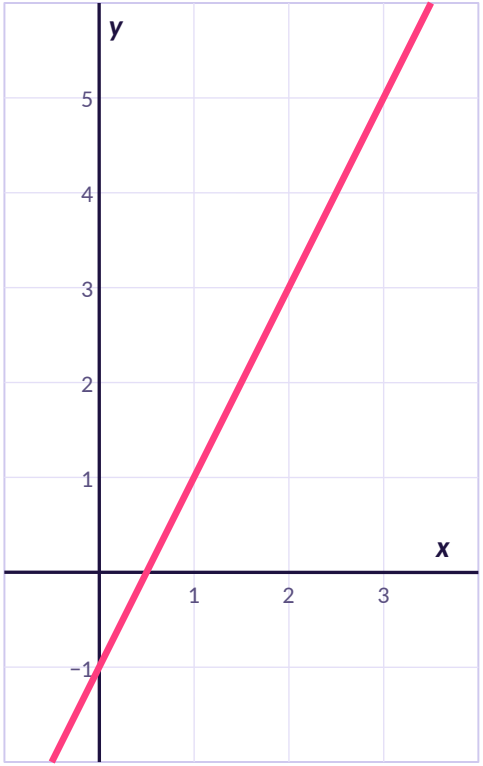
A ramp rises 1.5 m over 6 m across. Find its gradient. Is it steeper than a ramp with gradient 0.2?

1.5 ÷ 6 = 0.25, so yes it is steeper

Exam question 1

2 marks

Find the gradient of the line shown.



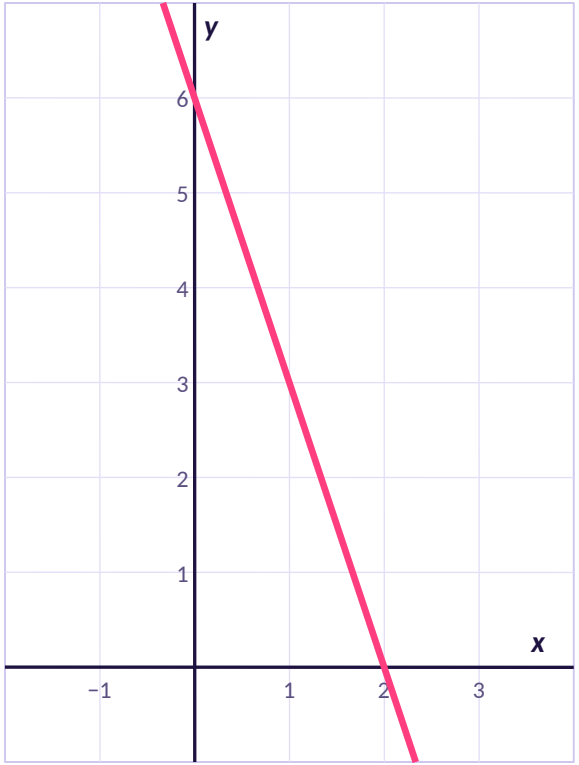
Space for working

Mark scheme: Up 6, across 3 (M1).
Gradient = 2 (A1)

Exam question 2

2 marks

The line shown goes down from left to right. Find its gradient.



Space for working

Mark scheme: Down 6, across 2 (M1).
Gradient = −3 (A1)

Objective 1 achieved

Find the gradient of a line from a graph



Objective 2

Find the gradient from two coordinates

✓ Achieved: Find the gradient of a line from a graph

Key idea

Learn

!

Gradient = change in y ÷ change in x.
Subtract in the same order for both. Take care with negatives: 3 – (–3) = 6.

Gradient =

$$\frac{y_2 - y_1}{x_2 - x_1}$$

Objective 2: Find the gradient from two coordinates

I do: example 1

I do

Find the gradient of the line through (1, 3) and (4, 9).

1. Change in y: $9 - 3 = 6$
2. Change in x: $4 - 1 = 3$
3. Gradient = $6 \div 3 = 2$

I do: example 2

I do

Find the gradient of the line through (2, 8) and (6, 6).

1. Change in y: $6 - 8 = -2$
2. Change in x: $6 - 2 = 4$
3. Gradient = $-2 \div 4 = -\frac{1}{2}$

I do: example 3

I do

Find the gradient of the line through (−1, 4) and (3, −8).

1. Change in y: −8 − 4 = −12
2. Change in x: 3 − (−1) = 4
3. Gradient = −12 ÷ 4 = −3

Mini whiteboards

We do

1. (0, 2) and (3, 11). Gradient?

Answer: 3

2. (1, 8) and (5, 0). Gradient?

Answer: −2

3. (2, 1) and (4, 9). Gradient?

Answer: 4

4. (−1, 3) and (3, 1). Gradient?

Answer: −½

You do: choose your challenge

You do

Bronze

Find the gradient of the line through (0, 1) and (2, 9).

Silver

Find the gradient of the line through (3, 10) and (5, 4).

Gold

Find the gradient of the line through (−2, −1) and (2, 1).

Platinum

The line through (1, 2) and (3, k) has gradient 4. Find k.

You do: answers

Check

Bronze

$8 \div 2 = 4$

Silver

$-6 \div 2 = -3$

Gold

$2 \div 4 = \frac{1}{2}$

Platinum

$(k - 2) \div 2 = 4$, so $k = 10$

If you finish early

Extra

Bronze

Find the gradient of the line through (0, 0) and (3, 12).

4

Silver

Find the gradient of the line through (1, 7) and (4, 1).

−2

Gold

Find the gradient of the line through (−3, 2) and (1, 4).

½

Platinum

The line through (2, k) and (5, 13) has gradient 3. Find k.

(13 − k) ÷ 3 = 3, so k = 4

Exam question 1

2 marks

A line passes through A(2, 1) and B(6, 9). Work out the gradient of the line AB.

Space for working

Mark scheme: 8 ÷ 4 (M1) = 2 (A1)

Exam question 2

2 marks

P is the point (−2, 5) and Q is the point (4, 2). Work out the gradient of PQ.

Space for working

Mark scheme: −3 ÷ 6 (M1) = −½ (A1)

Objective 2 achieved

Find the gradient from two coordinates



Real life

Apply

A wheelchair ramp rises 0.4 m over a horizontal distance of 6 m. The rules say its gradient must be less than 1/12. Is the ramp allowed?

Gradient = 0.4 ÷ 6 = 1/15 (about 0.067). 1/12 is about 0.083, so yes, the ramp is allowed

Extension: stretch yourself

Stretch

1. The line through (0, 3) and (4, k) has gradient −2. Find k.

(k − 3) ÷ 4 = −2, so k − 3 = −8 and k = −5

2. Are the lines through (0, 0) & (2, 6) and through (1, 1) & (3, 7) parallel?

Both have gradient 3, so yes, they are parallel

Exit ticket: exam questions

Exit

1. Find the gradient of the line through (0, 2) and (3, 11). **(2 marks)**

2. A line goes down 6 squares for every 3 squares across. What is its gradient? **(1 mark)**

3. Find the gradient of the line through (−2, 1) and (4, 4). **(2 marks)**

Exit ticket: mark scheme

Mark it

Q1 9 ÷ 3 = 3 [2]

Q2 −2 [1]

Q3 3 ÷ 6 = ½ [2]

Total: _____ / 5

Today I achieved

- ✔ **Find the gradient of a line from a graph**
- ✔ **Find the gradient from two coordinates**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Find the gradient of the line through (0, 0) and (2, 8). [1]

2. Find the gradient of the line through (1, 3) and (4, 12). [2]

3. Find the gradient of the line through (2, 9) and (6, 1). [2]

4. A line passes through (−1, 2) and (3, 4). Find its gradient. [2]

5. Line A goes through (0, 1) and (2, 7). Line B goes through (1, 0) and (4, 6). Which line is steeper? Show how you decide. [3]

Extra practice: Maths Genie → Edexcel question bank → Gradient of a Line